Conjugate fibres, certificate rigidity, and irrationality exponents

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Abstract

Let $\xi \in \mathbb{R}^\times$ and let $a, b \in \overline{\mathbb{Q}}^\times$ satisfy $e^{a\xi} = b$. We first prove two rigidity statements. Every algebraic exponential certificate for a fixed ξ has coefficient in one rational line, and every compatible embedding of the minimal certificate field is either the identity or a unique inversion involution. Thus, if $F = \mathbb{Q}(a, b)$ and $d = [F : \mathbb{Q}]$, there are at most two compatible embeddings. We prove

$$\mu(\xi) \le 2d,$$

improved to $\mu(\xi) \le d$ when the inversion involution exists. Equivalently, if $r \in \{1, 2\}$ is the number of compatible embeddings, then

$$\mu(\xi) \le \frac{2d}{r}.$$

The proof combines the bounded-fibre interpolation determinant of Liu–Jiang–Zhang with simultaneous number-field denominator clearing and the product formula. At the compatible places we jointly optimize the linear rational-approximation cost and the quadratic Taylor-collision energy. This removes the usual low/high transverse-index dichotomy and identifies $2d/r$ as the exact threshold of the resulting aggregate scalar estimate.

Rigidity reduces optimization over all certificates to power saturation of any one certificate. It produces an intrinsic integer $\kappa_{\exp}(\xi) \ge 1$, attained by a sufficiently divisible power, such that

$$\mu(\xi) \le 2\kappa_{\exp}(\xi).$$

We classify $\kappa_{\exp}(\xi) = 1$: it is precisely the union of split rational logarithms and logarithm coordinates of rational points on quadratic norm-one tori, including the completed quadratic arctanh / arctan family and its torsion boundary. Beyond complexity one we obtain, among other consequences, the optimal scalar bound $\mu(\log \alpha) \le 2d_{\mathrm{pow}}(\alpha)$ for positive algebraic α, relative norm-one bounds, and finite bounds for algebraic multiples of π and arguments of algebraic numbers.

1 Introduction

For an irrational real number x, its *irrationality exponent* is

$$\mu(x) = \sup\left\{\nu > 0 : 0 < \left|x - \frac{p}{q}\right| < q^{-\nu} \text{ for infinitely many } p \in \mathbb{Z},\ q \in \mathbb{Z}_{>0}\right\}.$$

Dirichlet's theorem gives $\mu(x) \ge 2$ for every irrational x; see, for example, [3, Chapter I]. Roth's theorem gives equality for every irrational algebraic number [12]. For explicit transcendental constants, proving exponent two is generally much harder.

OpenAI recently proved $\mu(\pi) = 2$ using a separated-weight interpolation determinant [11]. Two independent subsequent works moved the interpolation centres between fibres of the exponential coordinate. Brown proved the rational-logarithm theorem and supplied a Lean 4 formalization of that theorem; he also recorded in [2, Proposition 5.1] the bound $\mu(\log \alpha) \le 2[\mathbb{Q}(\alpha) : \mathbb{Q}]$ for positive real algebraic α. Liu, Jiang, and Zhang proved

$$\mu(\log q) = 2 \qquad (q \in \mathbb{Q}_{>0} \setminus \{1\})$$

and established the bounded-fibre moving-centre interpolation theorem that we use below [7]. Their paper explicitly isolates the next arithmetic obstruction: for an algebraic argument, clearing denominators produces a nonzero algebraic integer rather than a nonzero rational integer, so estimates are needed at its conjugate embeddings. The first purpose of this note is to carry out that comparison whenever some conjugate exponential fibres remain compatible.

Litsey recently converted the determinant proof of the irrationality of Catalan's constant into a finite irrationality measure by controlling mixed determinants under column perturbations and by retaining the energy cost of configurations with missing points [6, Theorem 5.1 and Proposition 5.10]. The compatible-place estimate below uses a related continuous-defect principle: rather than separating low and high transverse indices, it optimizes the approximation and collision penalties simultaneously. The argument itself is applied directly to the row-translation and collision bounds of Liu–Jiang–Zhang.

Eric Naslund publicly pointed out, in posts on X on 9 and 12 October 2026, the exact-two quadratic family

$$\frac{1}{\sqrt{D}} \operatorname{arctanh}(t\sqrt{D}), \qquad t \in \mathbb{Q},$$

with the appropriate real branches and torsion boundary; the later post linked a public Claude discussion [9]. For non-torsion members of this family, the moving-fibre interpolation supplied by Brown and Liu–Jiang–Zhang is needed in addition to the fixed-fibre argument for π. The new structural results here are the converse rigidity statement, the intrinsic power-saturation theorem, and the higher-degree bounds beyond that quadratic exact-two locus.

Fix $\xi \in \mathbb{R}^\times$ and $a, b \in \overline{\mathbb{Q}}^\times$ with $e^{a\xi} = b$. Let

$$F = \mathbb{Q}(a, b), \qquad d = [F : \mathbb{Q}], \qquad \Sigma_F = \operatorname{Hom}_{\mathbb{Q}}(F, \mathbb{C}),$$

and define

$$\Sigma_\xi(a, b) = \left\{ \sigma \in \Sigma_F : e^{\sigma(a)\xi} = \sigma(b) \right\}. \tag{1.1}$$

The distinguished inclusion belongs to $\Sigma_\xi(a, b)$, so $1 \le r := |\Sigma_\xi(a, b)| \le d$.

Theorem 1.1 (Rigid conjugate-fibre bound)**.** *Let $\xi \in \mathbb{R}^\times$ and $a, b \in \overline{\mathbb{Q}}^\times$ satisfy $e^{a\xi} = b$, and put $F = \mathbb{Q}(a, b)$ and $d = [F : \mathbb{Q}]$.*

(i) *One always has*

$$\boxed{\mu(\xi) \le 2d.}$$

(ii) *If F has an embedding $\tau : F \hookrightarrow \mathbb{C}$ satisfying*

$$\tau(a) = -a, \qquad \tau(b) = b^{-1},$$

then τ is a unique involution and

$$\boxed{\mu(\xi) \le d.}$$

Equivalently, if r denotes the number of compatible embeddings, then $r \in \{1, 2\}$ and

$$\boxed{\mu(\xi) \le \frac{2d}{r}.}$$

More precisely, for every real $\nu > 2d/r$ there is $Q = Q(a, b, \xi, \nu)$ such that

$$\left|\xi - \frac{p}{q}\right| \ge q^{-\nu} \qquad (p \in \mathbb{Z},\ q \in \mathbb{Z}_{>0},\ q \ge Q).$$

In particular, $\mu(\xi) = 2$ when $d = 1$, or when $d = 2$ and the inversion involution exists.

The exact-two assertions use Dirichlet's theorem and Hermite–Lindemann. If ξ were nonzero algebraic, then $a\xi$ would be nonzero algebraic, whereas $e^{a\xi} = b$ is algebraic, a contradiction; see [1, Chapter 1].

The second purpose of the note is to show that the scalar compatibility in Theorem 1.1 is rigid. If $e^{c\xi}$ is algebraic for another nonzero algebraic coefficient c, then $c/a \in \mathbb{Q}^\times$. Moreover, every compatible embedding of F sends

$$(a, b) \longmapsto (a, b) \quad \text{or} \quad (a, b) \longmapsto (-a, b^{-1}).$$

Thus $r \le 2$. This turns the optimized scalar theory into an integer-valued one: after replacing b by sufficiently divisible powers, the ratios d/r decrease to a positive integer $\kappa_{\exp}(\xi)$, and

$$\mu(\xi) \le 2\kappa_{\exp}(\xi).$$

The case $\kappa_{\exp}(\xi) = 1$ is classified by one-dimensional tori: the split torus gives rational logarithms, while a quadratic norm-one torus gives the arctanh / arctan family and its period boundary.

The paper is organized as follows. Section 2 proves the rigidity theorems. Sections 3–6 prove Theorem 1.1. Section 7 constructs the intrinsic integer complexity, and Section 8 classifies complexity one. Section 9 gives applications, and Section 10 records conjectural strengthenings.

2 Rigidity of algebraic exponential certificates

We use the Gelfond–Schneider theorem in the following standard form: if $\alpha \in \overline{\mathbb{Q}} \setminus \{0, 1\}$ and $\beta \in \overline{\mathbb{Q}} \setminus \mathbb{Q}$, then every value of α^β is transcendental; see [1, Chapter 1].

Theorem 2.1 (Certificate-line rigidity)**.** *Let $\xi \in \mathbb{R}^\times$, and suppose*

$$e^{a\xi} = b, \qquad e^{c\xi} = u, \qquad a, b, c, u \in \overline{\mathbb{Q}}^\times.$$

Then

$$\boxed{\frac{c}{a} \in \mathbb{Q}^\times.}$$

Thus the coefficients of all algebraic exponential certificates for a fixed ξ lie on one rational line.

Proof. Put $q = c/a \in \overline{\mathbb{Q}}^\times$. Suppose first that $b \ne 1$. Since $a\xi$ is a logarithm of b,

$$u = e^{c\xi} = e^{qa\xi}$$

is one value of b^q. If q were algebraic irrational, Gelfond–Schneider would make u transcendental. Hence $q \in \mathbb{Q}$.

Suppose now that $b = 1$. Since $a \neq 0$ and $\xi \neq 0$, there is a nonzero integer n such that

$$a\xi = 2\pi i n.$$

Then

$$u = e^{2\pi i n q}$$

is the value of $(-1)^{2nq}$ obtained from the logarithm $\log(-1) = \pi i$. If q were algebraic irrational, Gelfond–Schneider would again make u transcendental. Thus $q \in \mathbb{Q}^\times$ in all cases. □

Theorem 2.2 (Compatible-fibre rigidity)**.** *Let $e^{a\xi} = b$ and $F = \mathbb{Q}(a, b)$ as above. If $\sigma : F \hookrightarrow \mathbb{C}$ is compatible, then*

$$\boxed{(\sigma(a), \sigma(b)) = (a, b) \quad \text{or} \quad (\sigma(a), \sigma(b)) = (-a, b^{-1}).}$$

The first possibility is the distinguished inclusion. The second, if it occurs, is a unique involution in $\mathrm{Aut}(F/\mathbb{Q})$*. In particular,*

$$\boxed{r \in \{1, 2\}.}$$

Proof. Put $q = \sigma(a)/a$. If $b \neq 1$, compatibility gives

$$\sigma(b) = e^{\sigma(a)\xi} = e^{qa\xi},$$

so the same Gelfond–Schneider argument as in Theorem 2.1 gives $q \in \mathbb{Q}^\times$. If $b = 1$, write $a\xi = 2\pi i n$ and $\sigma(a)\xi = 2\pi i m$ with nonzero integers m, n; then $q = m/n \in \mathbb{Q}^\times$.

Extend σ to an automorphism g of a normal closure of F. Such an extension exists by the standard extension theorem for field embeddings; see, for example, [10, Chapter I]. If N is the finite order of g, then

$$a = g^N(a) = q^N a.$$

Hence $q^N = 1$. Since $q \in \mathbb{Q}^\times$, one has $q = 1$ or $q = -1$.

If $q = 1$, compatibility gives $\sigma(a) = a$ and $\sigma(b) = b$, so σ fixes the generators of F and is the distinguished inclusion. If $q = -1$, then

$$\sigma(a) = -a, \qquad \sigma(b) = e^{-a\xi} = b^{-1}.$$

These values on the generators determine at most one embedding. Its image is contained in F, so it is an automorphism of F; applying it twice fixes both generators, hence it is an involution. □

Remark 2.3 (Why a second compatible fibre occurs). Because ξ is real, complex conjugation is always a compatible embedding:

$$e^{\overline{a}\,\xi} = \overline{b}.$$

By Theorem 2.2, either $\overline{a} = a$ and $\overline{b} = b$, in which case the distinguished certificate field is real, or $\overline{a} = -a$ and $\overline{b} = b^{-1}$, in which case a is purely imaginary and $|b| = 1$. Thus nonreal certificates obtain their second compatible fibre from complex conjugation. For real certificates, a second fibre can only come from a real inversion involution; in the complexity-one case this is necessarily a real quadratic involution.

Corollary 2.4. *For every certificate, the ratio d/r is a positive integer. More precisely, if $r = 1$ then $d/r = d$, while if $r = 2$ and τ is the inversion involution, then*

$$\frac{d}{r} = \frac{d}{2} = [F^\tau : \mathbb{Q}].$$

Proof. Only the case $r = 2$ requires comment. Then τ is a nontrivial involution of F, so $[F : F^\tau] = 2$. □

3 The interpolation input

We recall the precise form of the geometric input that is needed. The following is [7, Theorem 9]. Fix positive rational weights

$$W = (w_0, w_1, \dots, w_m), \qquad V = (v_0, w_1/\theta, \dots, w_m/\theta), \qquad 0 < \theta < 1.$$

Let $\mathcal{P}_W(H)$ be the span of the monomials $Y^h X^\alpha$ with

$$w_0 h + \sum_{i=1}^{m} w_i \alpha_i \le H,$$

and let $\mathcal{J}_V(H)$ be the quotient of $\mathbb{C}[[t, u_1, \dots, u_m]]$ retaining precisely the monomials $t^s u^\beta$ with

$$v_0 s + \frac{1}{\theta} \sum_{i=1}^{m} w_i \beta_i < H.$$

Theorem 3.1 (Bounded-fibre moving-centre interpolation)**.** *Fix $m, K \ge 1$, $1 \le \varkappa \le K$, positive rational w_0, v_0, and $0 < \theta < 1$ such that*

$$K \left(\frac{w_0}{v_0} \right) \theta^m < 1, \qquad \varkappa \theta^m < 1. \tag{3.1}$$

There are successive lower thresholds for the positive rational weights $w_1, \dots, w_m$, independent of the interpolation centres, with the following property. Let

$$(y_j, c_{j1}, \dots, c_{jm}) \in \mathbb{C}^\times \times \mathbb{C}^m, \qquad 0 \le j < K,$$

assume that at most $\varkappa$ indices have a given value of y_j, and assume that

$$j \ne \ell,\ y_j = y_\ell \quad \Longrightarrow \quad c_{ji} \ne c_{\ell i} \quad (1 \le i \le m). \tag{3.2}$$

Then, for every sufficiently large admissible degree H, the map

$$\mathcal{P}_W(H) \longrightarrow \bigoplus_{j=0}^{K-1} \mathcal{J}_V(H)$$

obtained by substituting

$$Y = y_j(1 + t), \qquad X_i = c_{ji} + u_i + \log(1 + t)$$

is onto. The divisibility modulus for H depends only on the weights; the eventual threshold may depend on the centres.

The proof in [7] uses a local intersection-length estimate, separated weights, a curve inequality, an ordinary blow-up, and Serre vanishing. Brown independently proved a general-fibre version under the global condition $K\theta^m < 1$ [2, Theorem 2.1]. We quote the bounded-fibre formulation of Liu–Jiang–Zhang because its second volume constraint depends only on the maximal fibre multiplicity $\varkappa$, which is sharper when many multiplicative fibres are distinct and still covers torsion. We shall use the determinant translation and collision estimates from Sections 4–6 of Liu–Jiang–Zhang. Those estimates are uniform over the selected full-row minor and its columns.

3.1 Parameters

The optimized collision-energy estimate permits a particularly simple parameter choice.

Lemma 3.2 (Soft-gap parameters)**.** *Let d, r be positive integers and let $\nu > 2d/r$. There are rational numbers χ, θ, B, C satisfying*

$$0 < \chi < 1, \qquad \theta = \chi^2, \qquad C = \chi^{-1}, \qquad \theta < B < \chi, \tag{3.3}$$

and

$$g := r\nu(B - \theta) - d(1 - \theta) > 0. \tag{3.4}$$

In particular,

$$C\theta = \chi < 1, \qquad CB = \frac{B}{\chi} < 1, \qquad \frac{C\theta}{B} = \frac{\chi}{B} > 1, \qquad B^2 < \theta. \tag{3.5}$$

Proof. Since $r\nu > 2d$, one may choose a rational $\chi \in (0, 1)$ sufficiently close to 1 that

$$r\nu\chi > d(1 + \chi).$$

This is equivalent to

$$\chi^2 + \frac{d}{r\nu}(1 - \chi^2) < \chi.$$

Choose a rational number B strictly between the two sides, and put $\theta = \chi^2$ and $C = \chi^{-1}$. Then

$$r\nu(B - \theta) > d(1 - \theta),$$

which is (3.4); all statements in (3.5) are immediate. □

For a positive integer m, set

$$K = \lfloor C^m \rfloor, \qquad w_0 = B^{-m}, \qquad v_0 = 2K\theta^m w_0. \tag{3.6}$$

Then

$$K\left(\frac{w_0}{v_0}\right)\theta^m = \frac{1}{2}, \qquad K\theta^m \longrightarrow 0, \qquad \frac{K}{w_0} \longrightarrow 0, \qquad v_0 \longrightarrow \infty, \qquad \frac{m}{v_0} \longrightarrow 0. \tag{3.7}$$

Indeed, for all sufficiently large m,

$$K\theta^m \le (C\theta)^m = \chi^m, \qquad \frac{K}{w_0} \le (CB)^m = \left(\frac{B}{\chi}\right)^m,$$

while $K \ge C^m/2$ gives

$$v_0 \ge \left(\frac{C\theta}{B}\right)^m = \left(\frac{\chi}{B}\right)^m.$$

4 The number-field determinant

Fix the data of Theorem 1.1, and suppose for contradiction that there is a real $\nu > 2d/r$ for which

$$\left|\xi - \frac{p}{q}\right| \le q^{-\nu} \tag{4.1}$$

holds for arbitrarily large positive integers q. Replacing ν by a slightly smaller rational number still larger than $2d/r$, we may assume that ν is rational.

Choose the parameters of Lemma 3.2. Fix a rational truncation parameter

$$\Phi > \frac{2}{\theta}. \tag{4.2}$$

We shall first take m large, then choose m approximations

$$x_i = \frac{p_i}{q_i}, \qquad |\xi - x_i| \le q_i^{-\nu}, \qquad w_i = \lceil \log q_i \rceil, \qquad 1 \le i \le m,$$

successively. The denominators are taken large enough that $x_i \neq 0$, $|x_i| \le |\xi| + 1$, and the weights meet the successive thresholds in Theorem 3.1. This is possible because those thresholds are independent of the centres.

Use the centres

$$y_j = b^j, \qquad c_{ji} = jax_i, \qquad 0 \le j < K, \quad 1 \le i \le m. \tag{4.3}$$

We apply Theorem 3.1 with $\varkappa = K$. The second inequality in (3.1) holds for large m by $K\theta^m \to 0$. If $y_j = y_\ell$ and $j \neq \ell$, then

$$c_{ji} - c_{\ell i} = (j - \ell)ax_i \neq 0$$

for every i, so (3.2) holds. Thus the same argument covers roots of unity and non-torsion values of b.

Set

$$T_i = \left\lceil \frac{\Phi w_i}{v_0} \right\rceil, \qquad G_i(t) = \sum_{1 \le k < T_i} \frac{(-1)^{k+1}}{k} t^k, \qquad L_i = \operatorname{lcm}(1, \dots, T_i - 1), \tag{4.4}$$

where an empty sum and empty least common multiple are interpreted as 0 and 1. Because $v_0 T_i > w_i/\theta$, replacing $\log(1+t)$ by $G_i(t)$ is an invertible filtered change on every retained jet packet [7, Section 4].

For an admissible degree H, use the monomials

$$P = Y^h X^\alpha, \qquad w_0 h + \sum_i w_i \alpha_i \le H,$$

as columns, and use the triples

$$\varrho = (j, s, \beta), \qquad 0 \le j < K, \qquad v_0 s + \frac{1}{\theta} \sum_i w_i \beta_i < H,$$

as rows. The matrix entry is

$$[t^s u^\beta]\, b^{jh} (1+t)^h \prod_{i=1}^{m} (jax_i + G_i(t) + u_i)^{\alpha_i}. \tag{4.5}$$

For every sufficiently large admissible H, interpolation supplies a nonzero square minor $\Delta_H \in F^\times$ containing every row. Let $M = M_H$ be its size, and put

$$\bar{\beta}_H = \frac{1}{MH} \sum_{\varrho=(j,s,\beta)} \sum_i w_i \beta_i. \tag{4.6}$$

The row volume and its first transverse moment will both be useful.

Lemma 4.1 (Row volume and first moment)**.** *As $H \to \infty$ through admissible degrees,*

$$M \geq \frac{K\theta^m H^{m+1}}{(m+1)! v_0 \prod_i w_i}, \qquad M \sim \frac{K\theta^m H^{m+1}}{(m+1)! v_0 \prod_i w_i}, \tag{4.7}$$

and

$$\bar{\beta}_H = \frac{m}{m+2}\theta + O(H^{-1}), \qquad 0 \leq \bar{\beta}_H \leq \theta. \tag{4.8}$$

The implied constant depends only on the fixed weights.

Proof. Put

$$q_0 = v_0, \qquad q_i = \frac{w_i}{\theta} \quad (1 \leq i \leq m),$$

and let

$$\mathcal{R}_H = \left\{ n = (n_0, \dots, n_m) \in \mathbb{Z}_{\geq 0}^{m+1} : \sum_{i=0}^{m} q_i n_i < H \right\}.$$

The row set is the product of $\{0, \dots, K-1\}$ with $\mathcal{R}_H$. The unit cubes based at the points of $\mathcal{R}_H$ cover the simplex

$$\left\{ x \in \mathbb{R}_{\geq 0}^{m+1} : \sum_i q_i x_i < H \right\},$$

which gives the lower bound in (4.7). Comparing the same cubes with the two parallel simplices obtained by shifting the boundary by the fixed amount $\sum_i q_i$ gives

$$\#\mathcal{R}_H = \frac{H^{m+1}}{(m+1)! \prod_{i=0}^{m} q_i} + O(H^m).$$

The same comparison, now with the linear functions $q_i x_i$, gives

$$\sum_{n \in \mathcal{R}_H} q_i n_i = \frac{H^{m+2}}{(m+2)! \prod_{j=0}^{m} q_j} + O(H^{m+1}) \qquad (0 \leq i \leq m).$$

Since

$$\sum_i w_i \beta_i = \theta \sum_{i=1}^{m} q_i n_i,$$

dividing the corresponding first-moment asymptotic by $H\#\mathcal{R}_H$ gives $m\theta/(m+2) + O(H^{-1})$. Finally, every row satisfies $w\beta < \theta H$, which gives the stated upper bound. □

Remark 4.2 (Limiting row distribution)*.* Under uniform counting of rows, the normalized transverse weight $w\beta/(\theta H)$ converges to the Beta$(m, 2)$ distribution. Equation (4.8) is its first moment.

4.1 Algebraic-integral clearing and the norm inequality

Choose an integer $D_0 \geq 1$ such that

$$D_0 a,\ D_0 b \in \mathcal{O}_F.$$

Put

$$\mathfrak{D}_H = \prod_{i=1}^{m} L_i^{\lfloor H/w_i \rfloor}.$$

Scale the column (h, α) and the row (j, s, β) by

$$C_{h,\alpha} = D_0^{(K-1)h} \prod_i (D_0 q_i)^{\alpha_i}, \qquad S_{j,s,\beta} = \mathfrak{D}_H \prod_i (D_0 q_i)^{-\beta_i}. \tag{4.9}$$

Lemma 4.3 (Norm lower bound)**.** *Let* $\Lambda = 4 \log 2$ *and* $w_* = \min_i w_i$*. Define*

$$E_{\mathrm{ar}} = \frac{\Lambda \Phi m}{v_0} + \Lambda \sum_{i=1}^{m} \frac{1}{w_i} + \frac{\theta + \log D_0}{w_*} + \frac{(K-1)\log D_0}{w_0}. \tag{4.10}$$

Then

$$0 \leq \sum_{\sigma \in \Sigma_F} \frac{\log |\sigma(\Delta_H)|}{MH} + d(1 - \bar{\beta}_H + E_{\mathrm{ar}}). \tag{4.11}$$

Proof. The entry obtained from (4.5) after the row and column scalings is zero unless $\beta \leq \alpha$. Otherwise it equals

$$\mathfrak{D}_H (D_0 b)^{jh} D_0^{(K-1-j)h} \binom{\alpha}{\beta} [t^s](1+t)^h \cdot \prod_{i=1}^{m} \bigl(j(D_0 a) p_i + D_0 q_i G_i(t)\bigr)^{\alpha_i - \beta_i}.$$

The denominator contributed by the ith truncated logarithm divides $L_i^{\alpha_i - \beta_i}$, and $\alpha_i - \beta_i \leq \lfloor H/w_i \rfloor$. Hence every scaled entry belongs to $\mathcal{O}_F$. The scaled determinant

$$\widetilde{\Delta}_H = \Delta_H \prod_{\text{columns}} C_{h,\alpha} \prod_{\text{rows}} S_{j,s,\beta}$$

is therefore a nonzero algebraic integer. Consequently

$$1 \leq |\,\mathrm{N}_{F/\mathbb{Q}}(\widetilde{\Delta}_H)| = \prod_{\sigma \in \Sigma_F} |\sigma(\widetilde{\Delta}_H)|;$$

we count complex embeddings separately, as in the usual product formula [10, Chapter I].

It remains to bound the positive rational scaling factor. As in [7, Section 4],

$$\frac{\log \mathfrak{D}_H}{H} \leq \frac{\Lambda \Phi m}{v_0} + \Lambda \sum_i \frac{1}{w_i}.$$

Every selected column satisfies

$$\sum_i \alpha_i \log q_i \leq H, \qquad |\alpha| \leq \frac{H}{w_*}, \qquad h \leq \frac{H}{w_0}.$$

Moreover, since $0 \le w_i - \log q_i < 1$,

$$\sum_{\text{rows}} \sum_i \beta_i \log q_i \ge MH\bar{\beta}_H - \frac{MH\theta}{w_*}.$$

The negative row contribution involving $\log D_0$ may be discarded. Thus the logarithm of the total scaling factor, divided by MH, is at most $1 - \bar{\beta}_H + E_{\text{ar}}$. Taking logarithms of the norm inequality proves (4.11). □

5 Archimedean determinant estimates

Set

$$C_a = \max_{\sigma \in \Sigma_F} \max\{1, |\sigma(a)|\}, \qquad C_b = \max_{\sigma \in \Sigma_F} \max\{1, |\sigma(b)|\}, \qquad X = |\xi| + 1.$$

All constants in this section are uniform over the finite set Σ_F. The verification below is included to make explicit that neither the row translation nor the collision argument requires real centres or distinct multiplicative fibres. Reality enters only through the common rational approximants to ξ; repeated fibres are handled solely by the interpolation theorem that supplies the full-row minor.

5.1 Compatible embeddings

For $\sigma \in \Sigma_\xi(a, b)$, define

$$z_j(t) = j\sigma(a)\xi + \log(1+t), \qquad \epsilon_{ji} = j\sigma(a)(x_i - \xi), \qquad \tau_i(t) = G_i(t) - \log(1+t).$$

Compatibility gives the exact identities

$$e^{z_j(t)} = \sigma(b)^j (1+t), \qquad j\sigma(a)x_i + G_i(t) + u_i = z_j(t) + u_i + \epsilon_{ji} + \tau_i(t). \tag{5.1}$$

For a selected column $P = Y^h X^\alpha$ and a multi-index λ, let

$$f_{\lambda,P}(z) = [u^\lambda] P(e^z, z + u_1, \dots, z + u_m) = \binom{\alpha}{\lambda} e^{hz} z^{|\alpha| - |\lambda|},$$

with the convention that this is zero unless $\lambda \le \alpha$. Expanding (5.1) gives the same exact row translation as in [7, Section 5.1]:

$$\begin{aligned} \operatorname{row}^\sigma_{j,s,\beta} = \sum_{\substack{\lambda \ge \beta \\ w\lambda \le H}} \binom{\lambda}{\beta} \sum_{0 \le e \le \lambda - \beta} \binom{\lambda - \beta}{e} \epsilon_j^{\lambda - \beta - e} \\ \cdot \sum_{k=0}^{s} [t^k] \prod_i \tau_i(t)^{e_i} \left([t^{s-k}] f_{\lambda,P}(z_j(t)) \right)_P. \end{aligned} \tag{5.2}$$

Here and below $w\lambda = \sum_i w_i \lambda_i$.

Since

$$|\epsilon_{ji}| \le KC_a q_i^{-\nu} \le KC_a e^\nu e^{-\nu w_i},$$

and a nonzero tail coefficient in (5.2) uses $we < H/\Phi$, every scalar multiplying a translated test row is bounded by

$$\exp\{-\nu w(\lambda - \beta) + HE_{\text{tr}}\}, \tag{5.3}$$

where one may take

$$E_{\mathrm{tr}} = \frac{\nu}{\Phi} + \frac{\log 2}{v_0} + \frac{\log 4 + \log(2KC_a) + \nu}{w_*}. \tag{5.4}$$

Choose

$$R = 100K(1 + C_a|\xi|).$$

On $|t| \le 1/2$, every $z_j(t)$ lies in $|z| < R/2$. We use the Taylor-collision argument of [7, Section 5.2]; see also [5] for its interpolation-determinant form. Put

$$E_{\mathrm{hol}} = \frac{R}{w_0} + \frac{\log 2}{v_0} + \frac{\log(2R)}{w_*}, \qquad c = \frac{\log 2}{4}. \tag{5.5}$$

Let

$$S = \sum_i w_i.$$

The following elementary estimate is the soft replacement for the usual low/high transverse-index dichotomy.

Lemma 5.1 (Collision-energy optimization)**.** *Let $(n_\lambda)_{\lambda \in \mathbb{Z}^m_{\ge 0}}$ be finitely supported nonnegative real numbers with $\sum_\lambda n_\lambda = M$. For every $U > 0$,*

$$c\sum_\lambda n_\lambda^2 + \nu \sum_\lambda w\lambda\, n_\lambda \ge \nu UHM - \frac{\nu^2}{4c}\sum_\lambda (UH - w\lambda)_+^2. \tag{5.6}$$

Moreover,

$$\sum_{\lambda \in \mathbb{Z}^m_{\ge 0}} (UH - w\lambda)_+^2 \le \frac{2(UH + S)^{m+2}}{(m+2)! \prod_i w_i}. \tag{5.7}$$

Proof. For every $e \ge 0$ and $n \ge 0$, completion of the square gives

$$cn^2 + \nu e n \ge \nu UHn - \frac{\nu^2}{4c}(UH - e)_+^2.$$

Summing with $e = w\lambda$ proves (5.6). If $x \in \lambda + [0,1]^m$, then

$$UH + S - wx \ge UH - w\lambda.$$

Integration over the unit cubes therefore gives

$$\sum_\lambda (UH - w\lambda)_+^2 \le \int_{\mathbb{R}^m_{\ge 0}} (UH + S - wx)_+^2\, dx.$$

After the change of variables $y_i = w_i x_i$, the last integral equals the right side of (5.7). □

Lemma 5.2 (Optimized compatible-place estimate)**.** *There is a nonnegative quantity $\delta_H \to 0$, independent of $\sigma \in \Sigma_\xi(a,b)$ and of the selected full-row minor, such that for every $U > 0$,*

$$\begin{aligned} \frac{\log|\sigma(\Delta_H)|}{MH} \le{}& E_{\mathrm{tr}} + E_{\mathrm{hol}} + \delta_H - \nu(U - \bar\beta_H) \\ &+ \frac{\nu^2 v_0}{2cK\theta^m(m+2)} \left(U + \frac{S}{H}\right)^{m+2}. \end{aligned} \tag{5.8}$$

Define

$$U_m = \left(\frac{2cK\theta^m}{\nu v_0}\right)^{1/(m+1)}, \qquad \alpha_m = \frac{m+1}{m+2}U_m. \tag{5.9}$$

Then, for all sufficiently large H,

$$\frac{\log|\sigma(\Delta_H)|}{MH} \le E_{\mathrm{tr}} + E_{\mathrm{hol}} + \delta_H + \frac{\nu S}{H} - \nu(\alpha_m - \bar{\beta}_H). \tag{5.10}$$

For the parameters in (3.6),

$$U_m = \left(\frac{cB^m}{\nu}\right)^{1/(m+1)}, \qquad \alpha_m \longrightarrow B. \tag{5.11}$$

Proof. The row expansion (5.2) has only polynomially many terms in H for each row. Choose one translated row from each original row and let n_λ be the number chosen with transverse index λ. Expanding the entire functions $f_{\lambda,P}$ in Taylor series in z/R, any nonzero determinant contribution uses distinct Taylor degrees within each fixed λ-group. Hence

$$|\det T| \le \exp\left\{-c\sum_\lambda n_\lambda^2 + MH(E_{\mathrm{hol}} + o(1))\right\}. \tag{5.12}$$

To see explicitly why no reality or distinct-fibre hypothesis is hidden here, write

$$f_{\lambda,P}(z) = \sum_{d\ge 0} c_{\lambda,d,P}(z/R)^d.$$

The coefficient vector $(c_{\lambda,d,P})_P$ is independent of the centre index j and of the row order s; all dependence on the complex point $z_j(t)$ occurs in a scalar coefficient. Hence two selected rows with the same pair (λ, d) are proportional in the corresponding determinant term, which is the sole source of the collision saving. Cauchy's estimate is taken on the complex disk $|z| \le R$, so the argument is indifferent to whether the points $j\sigma(a)\xi$ are real or imaginary. It also does not require distinct multiplicative fibres. Indeed, if $\sigma(b)^j = \sigma(b)^\ell$ with $j \ne \ell$, then

$$z_j(0) - z_\ell(0) = (j-\ell)\sigma(a)\xi \ne 0,$$

although this additive distinctness is not used in the determinant cancellation. This is the collision estimate of [7, Section 5.2]; its proof is uniform in the columns and depends on the centres only through the radius R.

Multiplying the row scalars in (5.3) and using

$$\sum_{\mathrm{rows}} w\beta = MH\bar{\beta}_H$$

shows that the chosen determinant contribution is at most

$$\exp\{MH(E_{\mathrm{tr}} + E_{\mathrm{hol}} + o(1)) + \nu MH\bar{\beta}_H - \nu\sum_\lambda w\lambda\, n_\lambda - c\sum_\lambda n_\lambda^2\}.$$

Apply Lemma 5.1, then use the lower bound for M in (4.7). This gives (5.8). The total number of multilinear expansion terms is polynomial in H to the power M, so its logarithm is $o(MH)$ and is absorbed into δ_H.

For large H, take $U = U_m - S/H > 0$. The defining equation for U_m gives

$$\frac{\nu^2 v_0}{2cK\theta^m(m+2)} U_m^{m+2} = \frac{\nu U_m}{m+2}.$$

Substitution proves (5.10). Finally, $v_0 = 2K\theta^m w_0$ and $w_0 = B^{-m}$ give (5.11). □

Corollary 5.3 (Full-Plücker analytic bound)**.** *Let $\mathbf{P}_H^\sigma$ be the vector of all maximal minors of the conjugate interpolation matrix at $\sigma \in \Sigma_\xi(a,b)$. For every fixed $1 \le p \le \infty$ there is a nonnegative quantity $\delta_{H,p} \to 0$, uniform in $\sigma \in \Sigma_\xi(a,b)$, such that*

$$\frac{1}{MH} \log \|\mathbf{P}_H^\sigma\|_{\ell^p} \le E_{\mathrm{tr}} + E_{\mathrm{hol}} + \delta_{H,p} + \frac{\nu S}{H} - \nu(\alpha_m - \bar{\beta}_H). \tag{5.13}$$

Proof. The preceding proof is uniform in the selected maximal minor. If $N_H = \dim \mathcal{P}_W(H)$, then the number of maximal minors is at most 2^{N_H}. Here $N_H = O(H^{m+1})$, while $MH \asymp H^{m+2}$, so passing from the largest coordinate to any fixed ℓ^p-norm costs $o(MH)$ in the logarithm. □

Remark 5.4 (Water-filling profile and stability). The proof retains an exact stability remainder. For $w\lambda < UH$,

$$\begin{aligned} cn_\lambda^2 + \nu w\lambda\, n_\lambda - \nu UH n_\lambda + \frac{\nu^2}{4c}(UH - w\lambda)^2 \\ = c\left(n_\lambda - \frac{\nu}{2c}(UH - w\lambda)\right)^2. \end{aligned}$$

Thus a contribution close to the optimized bound must have occupancy close to the truncated affine profile

$$n_\lambda \approx \frac{\nu}{2c}(UH - w\lambda)_+.$$

This identifies precisely what additional incidence or cancellation information would be required to improve the aggregate estimate.

Remark 5.5 (Robustness of the collision constant). The precise value $c = \log 2/4$ is irrelevant to the limiting threshold. If the collision estimate in auxiliary dimension m has coefficient $c_m > 0$ and $\log c_m = o(m)$, then

$$U_m = B \exp\left(\frac{\log(c_m/(\nu B))}{m+1}\right) \longrightarrow B.$$

Thus any positive collision saving whose deterioration is subexponential in m leads to the same scalar threshold.

Remark 5.6 (Continuum compatibility of the incidence condition). At the continuum level, the bare condition $\lambda \ge \beta$ is not expected to create a leading-order gain. After summing over the row order s, the scaled row density on the θ-simplex is proportional to $\theta - \sum_i x_i$, while the water-filling density on the U-simplex is proportional to $U - \sum_i y_i$. The radial dilation $y = (U/\theta)x$ carries one profile to a constant multiple of the other and satisfies $y_i \ge x_i$ when $U > \theta$. Any further improvement must therefore use the detailed column incidence, determinant cancellation, or a non-scalar arithmetic input.

5.2 The remaining embeddings

At an incompatible embedding no approximation gain is available, but none is needed. A direct Cauchy estimate gives a normalized cost that can be made arbitrarily small by the separated parameter choices.

Lemma 5.7 (Crude estimate at every embedding)**.** *For every $\sigma \in \Sigma_F$,*

$$\frac{\log|\sigma(\Delta_H)|}{MH} \le E_{\mathrm{bad}} + \frac{\log M}{2H}, \tag{5.14}$$

where

$$E_{\mathrm{bad}} = \frac{K\log C_b + \log(3/2)}{w_0} + \frac{\log\bigl(2(1+KC_aX+\log 2)\bigr)}{w_*} + \frac{\log 2}{v_0}. \tag{5.15}$$

Proof. The conjugate of (4.5) is zero unless $\beta \le \alpha$. Otherwise it is

$$\sigma(b)^{jh}\binom{\alpha}{\beta}[t^s](1+t)^h \prod_i (j\sigma(a)x_i + G_i(t))^{\alpha_i-\beta_i}.$$

On $|t| = 1/2$ we have $|G_i(t)| \le \log 2$, while $|x_i| \le X$, $j < K$, $s \le H/v_0$, $h \le H/w_0$, and $|\alpha| \le H/w_*$. Cauchy's estimate and $\binom{\alpha}{\beta} \le 2^{|\alpha|}$ therefore bound each entry by $\exp(HE_{\mathrm{bad}})$. Hadamard's inequality contributes at most $M^{M/2}$, which gives (5.14). □

5.3 Making the errors small

The choices are made in the following order. First choose Φ so large that the contribution $r\nu/\Phi$ is smaller than any prescribed positive tolerance. Next let m tend to infinity. By (3.7) and (5.11),

$$\frac{m}{v_0} \to 0, \qquad \frac{K}{w_0} \to 0, \qquad \frac{1}{v_0} \to 0, \qquad \alpha_m \to B.$$

The constants D_0, C_a, C_b, ξ, d, r are fixed. Hence m may be chosen so that all terms in (4.10), (5.5), and (5.15) not involving w_i are as small as desired and, in addition,

$$r\nu(\alpha_m - \theta) - d(1-\theta) > \frac{g}{2}. \tag{5.16}$$

Finally, because the interpolation thresholds are centre-independent, the approximations may be selected successively with all w_i so large that

$$\sum_i \frac{1}{w_i}, \qquad \frac{1+\log K}{w_*}$$

are as small as desired. Once all parameters and centres are fixed, H tends to infinity through admissible degrees; then $\delta_H \to 0$, $S/H \to 0$, and $(\log M)/H \to 0$.

6 Proof of the conjugate-fibre theorem

Apply Lemma 5.2 at the r embeddings in $\Sigma_\xi(a,b)$ and Lemma 5.7 at the other $d-r$ embeddings. Substitution in (4.11) gives

$$0 \le d(1-\bar\beta_H) - r\nu(\alpha_m - \bar\beta_H) + \mathcal{E}_H, \tag{6.1}$$

where $\mathcal{E}_H$ is nonnegative. More explicitly, it is bounded by the corresponding linear combination of E_{ar}, E_{tr}, E_{hol}, E_{bad}, δ_H, $\nu S/H$, and $(\log M)/H$. The fixed-parameter terms can be made arbitrarily small by the preceding choices, and the remaining terms tend to zero with H.

Since $r\nu > d$ and $0 \le \bar{\beta}_H \le \theta$,

$$\begin{aligned} d(1-\bar{\beta}_H) - r\nu(\alpha_m - \bar{\beta}_H) &\le d(1-\theta) - r\nu(\alpha_m - \theta) \\ &< -\frac{g}{2} \end{aligned}$$

by (5.16). Taking the fixed errors sufficiently small and then H sufficiently large contradicts (6.1).

Therefore (4.1) cannot hold for arbitrarily large denominators when $\nu > 2d/r$. This proves the asserted eventual lower bound and hence $\mu(\xi) \le 2d/r$. As noted in the introduction, ξ is transcendental. When $d = r$—which by rigidity means either $d = r = 1$ or $d = r = 2$—Dirichlet's theorem supplies the opposite inequality $\mu(\xi) \ge 2$. This completes the proof of Theorem 1.1.

Proposition 6.1 (Finite-dimensional barrier of the aggregate estimate)**.** *Fix $m \ge 1$, $\nu \ge 2$, and parameters B, θ with $0 < B < 1$ and $B^2 < \theta < 1$, and put $c = \log 2/4$. Define*

$$U_m = \left(\frac{cB^m}{\nu}\right)^{1/(m+1)}, \qquad \alpha_m = \frac{m+1}{m+2}U_m, \qquad \beta_m = \frac{m}{m+2}\theta.$$

Then

$$2\alpha_m < 1 + \beta_m. \tag{6.2}$$

Consequently, if $r\nu \le 2d$, the limiting leading term supplied by the optimized aggregate estimate satisfies

$$d(1-\beta_m) - r\nu(\alpha_m - \beta_m) \ge 0. \tag{6.3}$$

Proof. Since $c = \log 2/4 < \nu$, one has

$$U_m < B^{m/(m+1)} < \theta^{m/(2m+2)}.$$

Set $x = \theta^{1/(2m+2)} \in (0,1)$. Then

$$\alpha_m < \frac{m+1}{m+2}x^m, \qquad \beta_m = \frac{m}{m+2}x^{2m+2}.$$

The function

$$f(x) = mx^{2m+2} - 2(m+1)x^m + m + 2$$

satisfies $f(1) = 0$ and

$$f'(x) = 2m(m+1)x^{m-1}(x^{m+2} - 1) \le 0 \qquad (0 < x \le 1).$$

Hence $f(x) \ge 0$, which proves (6.2). If $\alpha_m \le \beta_m$, then the left side of (6.3) is positive. If $\alpha_m > \beta_m$, then $r\nu \le 2d$ and (6.2) give

$$d(1-\beta_m) - r\nu(\alpha_m - \beta_m) \ge d(1 + \beta_m - 2\alpha_m) \ge 0.$$

□

Remark 6.2 (Why the scalar threshold is $2d/r$). By Lemma 4.1, $\bar{\beta}_H \to \beta_m$. Thus Proposition 6.1 shows that the fully optimized combination of the linear approximation cost and quadratic collision energy cannot, for any fixed auxiliary dimension, force a contradiction at or below $2d/r$. Conversely, Lemma 3.2 produces admissible parameters for every strict inequality $\nu > 2d/r$. In the limiting parameter geometry, B may approach $\sqrt{\theta}$, and

$$\frac{1-\theta}{\sqrt{\theta}-\theta} = 1 + \frac{1}{\sqrt{\theta}} \longrightarrow 2 \qquad (\theta \uparrow 1).$$

Thus the factor 2 comes from the interpolation geometry $B^2 < \theta$, while the factor d/r comes from the scalar product-formula comparison across conjugate places. Improvements below $2d/r$ must use information absent from the aggregate occupancy bound, such as detailed incidence, cancellation among minors, finite-place structure, or full-Plücker arithmetic.

Remark 6.3 (Effectivity). The theorem is qualitative. As in [7], after the centres have been selected the analytic contradiction has an explicit large-degree threshold, but the centre-dependent Serre-vanishing threshold in the interpolation theorem is not made effective.

7 Power saturation and intrinsic complexity

Fix one certificate $e^{a\xi} = b$. For $N \geq 1$, put

$$F_N = \mathbb{Q}(a, b^N), \qquad d_N = [F_N : \mathbb{Q}], \tag{7.1}$$

and let

$$r_N = \#\left\{\sigma : F_N \hookrightarrow \mathbb{C} : e^{N\sigma(a)\xi} = \sigma(b^N)\right\}. \tag{7.2}$$

Define

$$\kappa_N = \frac{d_N}{r_N}.$$

By Theorem 2.2 and Corollary 2.4, each r_N is one or two and each κ_N is a positive integer.

Lemma 7.1 (Divisibility monotonicity)**.** *If $N \mid M$, then*

$$\boxed{\kappa_M \leq \kappa_N.}$$

Proof. Since $N \mid M$, one has $F_M \subseteq F_N$. Every compatible embedding of F_N restricts to a compatible embedding of F_M, because the relation for N can be raised to the power M/N. Each embedding of F_M has at most $[F_N : F_M]$ extensions to F_N. Hence

$$r_N \leq [F_N : F_M] r_M.$$

Together with $d_N = [F_N : F_M] d_M$, this gives $d_N/r_N \geq d_M/r_M$. □

For an arbitrary certificate $e^{c\xi} = u$, write

$$\kappa(c, u; \xi) = \frac{[\mathbb{Q}(c, u) : \mathbb{Q}]}{\#\{\sigma : \mathbb{Q}(c, u) \hookrightarrow \mathbb{C} : e^{\sigma(c)\xi} = \sigma(u)\}}.$$

Lemma 7.2 (Rational roots do not improve the ratio)**.** *Let $e^{c\xi} = u$ be any algebraic exponential certificate. Then there is a positive integer M such that*

$$\boxed{\kappa(c, u; \xi) \geq \kappa_M.}$$

Proof. By Theorem 2.1, $c = qa$ for some $q \in \mathbb{Q}^\times$. Replacing (c, u) by $(-c, u^{-1})$ if necessary does not change its field or compatible-embedding count, so write $q = m/n$ with coprime positive integers m, n. Then

$$u^n = b^m.$$

Put

$$F' = \mathbb{Q}(a, u), \qquad F_m = \mathbb{Q}(a, b^m) \subseteq F',$$

and let r' be the number of compatible embeddings of F' for the certificate $e^{c\xi} = u$. Every such embedding restricts to a compatible embedding of F_m: raising $e^{(m/n)\sigma(a)\xi} = \sigma(u)$ to the nth power gives $e^{m\sigma(a)\xi} = \sigma(b^m)$. Therefore

$$r' \le [F' : F_m] r_m.$$

Since $[F' : \mathbb{Q}] = [F' : F_m] d_m$, one obtains

$$\kappa(c, u; \xi) = \frac{[F' : \mathbb{Q}]}{r'} \ge \frac{d_m}{r_m} = \kappa_m.$$

□

Definition 7.3 (Exponential certificate complexity)**.** For a real nonzero ξ admitting an algebraic exponential certificate, set

$$\kappa_{\exp}(\xi) = \inf_{e^{c\xi} = u} \kappa(c, u; \xi),$$

where the infimum runs over all $c, u \in \overline{\mathbb{Q}}^\times$.

Theorem 7.4 (Intrinsic power saturation)**.** *For any fixed certificate $e^{a\xi} = b$,*

$$\boxed{\kappa_{\exp}(\xi) = \min_{N \ge 1} \kappa_N = \lim_{j \to \infty} \kappa_{j!} \in \mathbb{Z}_{>0}.}$$

The sequence $\kappa_{j!}$ is eventually constant. In particular, the optimum is attained and is independent of the initial certificate. Moreover,

$$\boxed{\mu(\xi) \le 2\kappa_{\exp}(\xi).}$$

Proof. Powers of the fixed certificate are among all certificates, so $\kappa_{\exp}(\xi) \le \inf_N \kappa_N$. Conversely, Lemma 7.2 shows that every certificate has ratio at least one of the κ_N. Hence equality holds.

By Lemma 7.1, the positive integers $\kappa_{j!}$ form a nonincreasing sequence and are therefore eventually constant. For every fixed N, one has $N \mid j!$ for all sufficiently large j, so the eventual value is at most κ_N. It is therefore the global minimum. Finally, apply Theorem 1.1 to a power attaining this minimum. □

If one defines the optimized compatible-fibre density by

$$\rho_{\exp}(\xi) = \sup_{e^{c\xi} = u} \frac{r(c, u; \xi)}{[\mathbb{Q}(c, u) : \mathbb{Q}]},$$

then Theorem 7.4 says

$$\boxed{\rho_{\exp}(\xi) = \frac{1}{\kappa_{\exp}(\xi)}.} \tag{7.3}$$

Thus the only possible optimized densities are $1, 1/2, 1/3, \ldots$.

Corollary 7.5 (Torsion-defect criterion)**.** *Let L be a normal closure of $\mathbb{Q}(a,b)$. Suppose that for every embedding $\tau : L \hookrightarrow \mathbb{C}$ the defect*

$$\delta_\tau = \frac{e^{\tau(a)\xi}}{\tau(b)}$$

is a root of unity. Then

$$\boxed{\kappa_{\exp}(\xi) = 1, \qquad \mu(\xi) = 2.}$$

Proof. Choose N divisible by the orders of the finitely many roots of unity δ_τ. Then

$$e^{N\tau(a)\xi} = \tau(b^N)$$

for every τ. Every embedding of $F_N = \mathbb{Q}(a, b^N)$ extends to an embedding of L, so every embedding of F_N is compatible. Hence $\kappa_N = 1$ and Theorem 7.4 applies. □

Remark 7.6 (Relation lattices). Suppose $e^{a_j\xi} = b_j$ for finitely many algebraic certificates. By Theorem 2.1, all a_j are rational multiples of one coefficient. Any integer combination of the certificates is therefore another rational rescaling, and Lemma 7.2 shows that it cannot improve on power saturation. Thus no separate relation-lattice optimization is needed.

8 Exact scalar complexity one and one-dimensional tori

The point of the next theorem is a converse: Gelfond–Schneider rigidity shows that, after rational scaling, branch completion, and the torsion boundary, no other exact-two family can be reached by the scalar compatible-fibre method.

Theorem 8.1 (Classification of $\kappa_{\exp} = 1$)**.** *Let $\xi \in \mathbb{R}^\times$ admit an algebraic exponential certificate. The following are equivalent.*

(i) $\kappa_{\exp}(\xi) = 1$.

(ii) *Some powered certificate has all of its embeddings compatible.*

(iii) *One of the following alternatives holds.*

(a) *There are $\lambda_0 \in \mathbb{Q}^\times$ and $\alpha_0 \in \mathbb{Q}_{>0} \setminus \{1\}$ such that*

$$\xi = \lambda_0 \log \alpha_0.$$

(b) *There are a quadratic field $E = \mathbb{Q}(\omega)$ with $\omega^2 \in \mathbb{Q}^\times$, its nontrivial automorphism $\tau(\omega) = -\omega$, and an element $B \in E^\times$ satisfying*

$$\tau(B) = B^{-1}, \qquad e^{2\omega\xi} = B.$$

In alternative (iii)(b)*, if $B \neq -1$ then*

$$t = \frac{B-1}{\omega(B+1)} \in \mathbb{Q}$$

and, for some $k \in \mathbb{Z}$ and a compatible branch of arctanh*,*

$$\boxed{\xi = \frac{1}{\omega}\operatorname{arctanh}(t\omega) + \frac{\pi i k}{\omega}.} \tag{8.1}$$

If $B = -1$, then

$$\boxed{\xi = \frac{(2k+1)\pi i}{2\omega}} \qquad (k \in \mathbb{Z}). \tag{8.2}$$

Every number in the theorem has irrationality exponent two.

Proof. The equivalence of (i) and (ii) follows from the attainment statement in Theorem 7.4. Assume (ii), and choose a powered certificate

$$e^{A\xi} = B, \qquad F = \mathbb{Q}(A, B),$$

with every embedding compatible. By Theorem 2.2, $[F : \mathbb{Q}] \le 2$.

If $F = \mathbb{Q}$, then $A, B \in \mathbb{Q}^\times$. Since A and ξ are real, $B = e^{A\xi}$ is positive, and $B \neq 1$. Thus

$$\xi = A^{-1} \log B,$$

which is alternative (iii)(a).

Suppose $[F : \mathbb{Q}] = 2$, and let τ be its nontrivial automorphism. Compatibility and Theorem 2.2 give

$$\tau(A) = -A, \qquad \tau(B) = B^{-1}.$$

Put $\omega = A/2$. Then $\omega \notin \mathbb{Q}$, $F = \mathbb{Q}(\omega)$, and $\omega^2 \in \mathbb{Q}^\times$. This is alternative (iii)(b).

If $B \neq -1$, define t by the displayed formula. Direct calculation using $\tau(\omega) = -\omega$ and $\tau(B) = B^{-1}$ gives $\tau(t) = t$, hence $t \in \mathbb{Q}$. Solving for B yields

$$B = \frac{1 + t\omega}{1 - t\omega}.$$

Since $e^{2\omega\xi} = B$, the logarithms of the two sides differ by $2\pi ik$, which gives (8.1). If $B = -1$, then $2\omega\xi = (2k+1)\pi i$, giving (8.2).

Conversely, alternative (iii)(a) gives the rational certificate $e^{\xi/\lambda_0} = \alpha_0$. In alternative (iii)(b), the identity and τ are both compatible with the certificate $e^{2\omega\xi} = B$. Hence $\kappa_{\exp}(\xi) = 1$ in either case. The exponent-two conclusion now follows from Theorem 7.4 and Dirichlet's theorem. □

Remark 8.2 (Torus interpretation). Alternative (iii)(a) is the split torus $\mathbb{G}_m$. Alternative (iii)(b) is the quadratic norm-one torus $\mathrm{Res}^1_{E/\mathbb{Q}} \mathbb{G}_m$; the parameter $B = (1 + t\omega)/(1 - t\omega)$ is its Cayley parametrization away from $B = -1$. Thus scalar complexity one is exactly the locus of nonzero real logarithm coordinates of $\mathbb{Q}$-rational points on one-dimensional $\mathbb{Q}$-tori. For general background on algebraic tori, see [8, Chapter 12].

Corollary 8.3 (Quadratic artanh family)**.** *Let $D \in \mathbb{Q}^\times$, let $t \in \mathbb{Q}$ with $1 - t^2 D \neq 0$, and choose a branch for which*

$$\xi = \frac{1}{\sqrt{D}} \operatorname{arctanh}(t\sqrt{D}) + \frac{\pi i k}{\sqrt{D}}$$

is real and nonzero. Then

$$\boxed{\mu(\xi) = 2.}$$

The torsion boundary

$$\xi = \frac{(2k+1)\pi i}{2\sqrt{D}}$$

is included as well whenever it is real and nonzero.

Proof. If $\sqrt{D} \in \mathbb{Q}$, the value is a rational multiple of a logarithm of a positive rational number. Otherwise apply Theorem 8.1 with $\omega = \sqrt{D}$ and $B = (1 + t\sqrt{D})/(1 - t\sqrt{D})$. □

For $D > 0$, the principal real branch has $k = 0$ and $|t|\sqrt{D} < 1$. Taking $D = 1$ and $t = (\alpha_0 - 1)/(\alpha_0 + 1)$ shows that, up to the rational factor 2, the family contains every $\log \alpha_0$ with $\alpha_0 \in \mathbb{Q}_{>0} \setminus \{1\}$. For $D = -s < 0$,

$$\frac{1}{\sqrt{D}} \operatorname{arctanh}(t\sqrt{D}) = \frac{1}{\sqrt{s}} \arctan(t\sqrt{s}),$$

so it contains rational arctangents. Its period boundary contains the square-root rescalings of π.

Corollary 8.4 (Concrete exact-two values)**.** *For every integer $k \geq 2$,*

$$\mu\left(\arctan \frac{1}{k}\right) = 2, \qquad \mu\left(\frac{\pi}{\sqrt{3}}\right) = 2,$$

and

$$\boxed{\mu\left(\sqrt{2k-1} \arctan \frac{\sqrt{2k-1}}{k-1}\right) = 2.}$$

Proof. The first assertion is the case $D = -1$, $t = 1/k$. For the second, take $D = -3$, $t = 1$ and multiply the resulting number $\pi/(3\sqrt{3})$ by 3. For the last, put $s = 2k - 1$ and $t = 1/(k - 1)$ in the negative-D form above, then multiply by the rational number s. Irrationality exponent is invariant under nonzero rational scaling. □

Corollary 8.5 (Quadratic norm-one elements)**.** *Let $E = \mathbb{Q}(\sqrt{D})$ be real quadratic, and let $u \in E^\times$ satisfy*

$$u > 0, \qquad u \neq 1, \qquad \mathrm{N}_{E/\mathbb{Q}}(u) = 1.$$

Then

$$\boxed{\mu\left(\frac{\log u}{\sqrt{D}}\right) = 2.}$$

If $\mathrm{N}_{E/\mathbb{Q}}(u) = -1$, the same conclusion follows after squaring the certificate.

Proof. The nontrivial automorphism sends $\sqrt{D}$ to $-\sqrt{D}$ and u to u^{-1}. Apply Theorem 8.1; for norm -1, use u^2. □

9 Applications beyond complexity one

9.1 Positive algebraic logarithms

Brown independently obtained the degree bound $\mu(\log \alpha) \leq 2[\mathbb{Q}(\alpha) : \mathbb{Q}]$ for positive real algebraic α [2, Proposition 5.1]. The theorem below replaces the degree by the generally smaller power degree and proves that this is the best possible bound within the entire scalar certificate architecture. Liu–Jiang–Zhang already obtain exact exponent two when a positive power of α is rational [7, Corollary 7].

For $\alpha \in \overline{\mathbb{Q}}_{>0} \setminus \{1\}$, define

$$d_{\mathrm{pow}}(\alpha) = \min_{N \geq 1} [\mathbb{Q}(\alpha^N) : \mathbb{Q}]. \tag{9.1}$$

Theorem 9.1 (Exact scalar complexity of an algebraic logarithm)**.** *For every positive algebraic* $\alpha \neq 1$*,*

$$\boxed{\kappa_{\exp}(\log \alpha) = d_{\mathrm{pow}}(\alpha)}$$

and consequently

$$\boxed{\mu(\log \alpha) \leq 2d_{\mathrm{pow}}(\alpha).}$$

If some positive power of α *is rational, then* $\mu(\log \alpha) = 2$*.*

Proof. Use the initial certificate $e^{\log \alpha} = \alpha$. For its Nth power, the coefficient N is rational and $F_N = \mathbb{Q}(\alpha^N)$. A compatible embedding must satisfy

$$\sigma(\alpha^N) = e^{N \log \alpha} = \alpha^N,$$

so it fixes the generator of F_N and is the identity. Thus $r_N = 1$ and $\kappa_N = [\mathbb{Q}(\alpha^N) : \mathbb{Q}]$. Apply Theorem 7.4. The rational-power case also follows directly from [7, Corollary 7]. □

Remark 9.2 (Algebraic rescaling asymmetry). The norm-one theorem gives

$$\mu\left(\frac{\log(3 + 2\sqrt{2})}{\sqrt{2}}\right) = 2,$$

whereas the scalar complexity of $\log(1 + \sqrt{2})$ is 2, so the present method gives only

$$\mu(\log(1 + \sqrt{2})) \leq 4.$$

Thus the scalar theory is not invariant under multiplication by a nonrational algebraic number. This sharp asymmetry is one motivation for the full-Plücker conjectural mechanism discussed in Section 10.

The power degree has a useful Galois description.

Proposition 9.3 (Conjugates modulo torsion)**.** *Let* L *be a normal closure of* $\mathbb{Q}(\alpha)$*, put* $G = \mathrm{Gal}(L/\mathbb{Q})$*, and define*

$$H_\alpha = \left\{ g \in G : \frac{g(\alpha)}{\alpha} \text{ is a root of unity} \right\}.$$

Then H_α *is a subgroup of* G *and*

$$\boxed{d_{\mathrm{pow}}(\alpha) = [G : H_\alpha].}$$

Proof. The root-of-unity condition is preserved under products and inverses, so H_α is a subgroup. Choose N divisible by the orders of all root-of-unity ratios $g(\alpha)/\alpha$ with $g \in H_\alpha$. Then H_α fixes α^N. Conversely, if $g(\alpha^N) = \alpha^N$, then $g(\alpha)/\alpha$ is an Nth root of unity, so $g \in H_\alpha$. Thus $H_\alpha = \mathrm{Gal}(L/\mathbb{Q}(\alpha^N))$, and $[\mathbb{Q}(\alpha^N) : \mathbb{Q}] = [G : H_\alpha]$. This value is minimal over all powers. □

9.2 Relative norm-one fibres

Theorem 9.4 (Relative norm-one bound)**.** *Let* E *be a number field, let* $\omega, t \in \overline{\mathbb{Q}}$ *satisfy*

$$E = \mathbb{Q}(\omega^2, t), \qquad \omega \notin E,$$

and assume $1 - t\omega \neq 0$*. Put*

$$u = \frac{1 + t\omega}{1 - t\omega}.$$

Suppose

$$\xi = \frac{1}{\omega}\operatorname{arctanh}(t\omega)$$

is real and nonzero for a chosen branch. Then, with $F = \mathbb{Q}(\omega, u)$,

$$\boxed{\mu(\xi) \le [F : \mathbb{Q}] \le 2[E : \mathbb{Q}].}$$

Proof. The extension $E(\omega)/E$ is quadratic with involution $\tau(\omega) = -\omega$. Since $t \in E$,

$$\tau(u) = u^{-1}.$$

The relation $e^{2\omega\xi} = u$ therefore has the identity and τ as compatible embeddings of its minimal certificate field F. Hence $r = 2$, and Theorem 1.1 gives $\mu(\xi) \le [F : \mathbb{Q}]$. Finally $F \subseteq E(\omega)$. □

The rational quadratic-artanh family is the case $E = \mathbb{Q}$. For example, with $\omega = \sqrt[4]{2}$ and rational t in the real domain,

$$\mu\left(\frac{1}{\sqrt[4]{2}}\operatorname{arctanh}(t\sqrt[4]{2})\right) \le 4.$$

9.3 Algebraic multiples of π

Corollary 9.5. *For every nonzero real algebraic number* λ,

$$\boxed{\mu(\lambda\pi) \le [\mathbb{Q}(i/\lambda) : \mathbb{Q}] \le 2[\mathbb{Q}(\lambda) : \mathbb{Q}].}$$

If $\lambda^2 \in \mathbb{Q}^\times$, *then* $\mu(\lambda\pi) = 2$. *For example,*

$$\mu(2^{1/3}\pi) \le 6.$$

Proof. Use the certificate

$$e^{(i/\lambda)(\lambda\pi)} = -1.$$

Complex conjugation fixes λ and sends i/λ to its negative, so the minimal field has two compatible embeddings. The first bound follows from Theorem 1.1; the second follows from $\mathbb{Q}(i/\lambda) \subseteq \mathbb{Q}(i, \lambda)$. If λ^2 is rational, then $\mathbb{Q}(i/\lambda)$ is quadratic and both embeddings are compatible. □

9.4 Arguments of algebraic numbers

Corollary 9.6. *Let* $z \in \overline{\mathbb{Q}}^\times$, *and let* $\vartheta \in \mathbb{R}^\times$ *be an argument of* z. *Put*

$$\beta = \frac{z}{\overline{z}}.$$

Then

$$\boxed{\mu(\vartheta) \le \min_{N \ge 1}[\mathbb{Q}(i, \beta^N) : \mathbb{Q}].}$$

In particular, for every nonzero real algebraic α,

$$\boxed{\mu(\arctan\alpha) \le 2[\mathbb{Q}(\alpha) : \mathbb{Q}].}$$

For rational $\alpha \neq 0$, *equality* $\mu(\arctan\alpha) = 2$ *holds.*

Proof. Since $e^{2i\vartheta} = \beta$, complex conjugation sends the coefficient $2i$ to $-2i$ and β to β^{-1}. The powered certificate therefore has two compatible embeddings in $\mathbb{Q}(i, \beta^N)$, so Theorem 1.1 gives the first bound. For $\vartheta = \arctan\alpha$, take $z = 1 + i\alpha$ and use $\mathbb{Q}(i, \beta) \subseteq \mathbb{Q}(i, \alpha)$. The rational case has field $\mathbb{Q}(i)$ and complexity one. □

9.5 Rational projective invariance

Proposition 9.7 ($\mathrm{PGL}_2(\mathbb{Q})$-invariance)**.** *Let x be irrational and let*

$$g(X) = \frac{AX + B}{CX + D} \in \mathrm{PGL}_2(\mathbb{Q})$$

be defined at x. Then

$$\boxed{\mu(g(x)) = \mu(x).}$$

Proof. After multiplying the coefficients by a common denominator, assume $A, B, C, D \in \mathbb{Z}$. For a rational p/q tending to x,

$$g(x) - g(p/q) = \frac{(AD - BC)(x - p/q)}{(Cx + D)(Cp/q + D)}.$$

The two denominator factors stay bounded above and away from zero. The rational number

$$g(p/q) = \frac{Ap + Bq}{Cp + Dq}$$

has reduced denominator $O(q)$. Hence approximations to x of every exponent ν yield approximations to $g(x)$ of every smaller exponent. Thus $\mu(g(x)) \geq \mu(x)$. Applying the same argument to g^{-1} gives the reverse inequality. □

Consequently every exact-two family above produces an entire rational Möbius orbit of exact-two constants, including reciprocals and rational affine transforms.

10 Conjectures and further directions

Rigidity shows that the scalar product-formula method is now essentially classified. When $\kappa_{\exp}(\xi) > 1$, no hidden collection of three or more compatible scalar fibres can improve the bound. Moreover, Proposition 6.1 shows that even the optimal aggregate use of the approximation and collision energies remains subject to the same threshold. Any general passage from $2\kappa_{\exp}(\xi)$ to 2 must therefore use a genuinely non-scalar arithmetic or incidence input.

Conjecture 10.1 (Algebraic exponential-fibre conjecture)**.** *If*

$$\xi \in \mathbb{R}^\times, \qquad a, b \in \overline{\mathbb{Q}}^\times, \qquad e^{a\xi} = b,$$

then

$$\boxed{\mu(\xi) = 2.}$$

The first benchmark beyond the scalar theory is

$$\mu(\log \alpha) = 2 \qquad (\alpha \in \overline{\mathbb{Q}}_{>0} \setminus \{1\}),$$

for which Theorem 9.1 currently gives the optimal scalar bound $2d_{\mathrm{pow}}(\alpha)$.

Conjecture 10.2 (Algebraic linear forms in logarithms)**.** *Let $\alpha_1, \dots, \alpha_s \in \overline{\mathbb{Q}}^\times$, choose logarithm branches, and let $\beta_1, \dots, \beta_s \in \overline{\mathbb{Q}}$. If*

$$\Lambda = \sum_{j=1}^{s} \beta_j \log \alpha_j$$

is real and nonzero, then

$$\boxed{\mu(\Lambda) = 2.}$$

Baker's theorem implies that such a nonzero Λ is transcendental [1, Chapter 2]; the conjecture asks for the optimal rational approximation exponent. It contains conjecture 10.1 as the one-logarithm case.

10.1 Full-Plücker arithmetic

A plausible route to conjecture 10.1 is to retain the entire vector of maximal minors of the interpolation matrix rather than one scalar minor. The scalar norm argument records only one Plücker coordinate and is forced to pay the integer complexity $\kappa_{\exp}$. In contrast, Corollary 5.3 proves that the analytic smallness estimate already holds uniformly for every maximal minor and hence for every fixed norm of the full Plücker vector. The remaining problem is arithmetic: one needs an adelic projective-height principle that converts simultaneous smallness of all maximal minors at the compatible places into an exceptional functional relation without paying separately for every incompatible scalar coordinate.

An adelic full-Plücker argument, potentially combined with the Subspace Theorem [13], could produce such a relation. The moving-centre interpolation theorem is designed to exclude precisely these functional dependencies. The mixed-determinant perturbation estimates of Litsey provide related evidence that families of minors can be controlled stably under structured perturbations [6, Sections 5.2–5.6]. Making the required projective product-formula mechanism quantitative remains open.

10.2 Near-compatible powers

Let L be a normal closure of $\mathbb{Q}(a, b)$ and, for $\tau : L \hookrightarrow \mathbb{C}$, put

$$\delta_\tau = \frac{e^{\tau(a)\xi}}{\tau(b)}.$$

The torsion-defect criterion Corollary 7.5 treats the case in which every δ_τ is a root of unity.

Conjecture 10.3 (Unitary-defect exact-two principle)**.** *If $|\delta_\tau| = 1$ for every embedding $\tau : L \hookrightarrow \mathbb{C}$, then*

$$\boxed{\mu(\xi) = 2.}$$

The root-of-unity case follows from power saturation. In the general unitary case, simultaneous recurrence can make finitely many powers δ_τ^N close to 1, but the determinant also amplifies the remaining mismatch through the parameters K and H. A proof would therefore require a quantitative near-compatible-fibre estimate together with a coupled choice of the power N and the interpolation scale. The fixed-certificate argument of the present paper does not provide such a coupling.

10.3 Effectivity of the geometric input

Conjecture 10.4 (Effective special-centre interpolation)**.** *For algebraic a, b, rational $x_1, \dots, x_m$, and centres*

$$y_j = b^j, \qquad c_{ji} = jax_i, \qquad 0 \le j < K,$$

the eventual surjectivity threshold in Theorem 3.1 is bounded effectively in terms of the weights, m, K, the algebraic degrees, and the logarithmic Weil heights of $a, b, x_1, \dots, x_m$.

Such a result would make the geometric part of the determinant argument quantitative and would be an essential input for near-compatible powers or explicit finite-place estimates. It would also isolate whether the remaining ineffectivity is intrinsic to the selection of rational approximants or only to the present use of Serre vanishing.

Acknowledgments and provenance

We thank Eric Naslund for publicly drawing attention, in posts on X on 9 and 12 October 2026, to the quadratic $\operatorname{arctanh}(t\sqrt{D})/\sqrt{D}$ family; the later post linked a public Claude discussion [9]. We cite those posts as prior public identification of the family. The converse rigidity, intrinsic power-saturation, and higher-degree bounds are contributions of the present paper.

The geometric interpolation theorem and the analytic row-translation and Taylor-collision estimates used here are those of Liu–Jiang–Zhang [7], building on the interpolation-determinant method of Laurent [4, 5] and the proof for π in [11]. Brown independently obtained the rational-logarithm theorem and a general-fibre interpolation theorem under a global fibre-count condition, together with a Lean 4 formalization of the rational theorem, and recorded the real-algebraic degree bound [2]. Litsey's mixed-determinant perturbation argument for Catalan's constant provides related contemporary context for continuous defect optimization [6]. The present paper combines the Liu–Jiang–Zhang approximation and collision penalties by the soft convex optimization of Lemma 5.1; the new arithmetic determinant step is the simultaneous number-field denominator clearing and product-formula comparison across compatible and incompatible conjugate fibres. The rigidity, intrinsic power-saturation, and torus classification are obtained by combining that comparison with Gelfond–Schneider.

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